

TOP Data Engineer Interview Preparation Questions

1) Write a query to find employees with the second highest salary in a table?

Answer:

To solve this, use a subquery to first find the maximum salary, then filter out that value to get the second highest

Example:

```
SELECT MAX(salary) AS second_highest_salary FROM employees  
WHERE salary < (SELECT MAX(salary) FROM employees);
```

Explanation:

. Inner query (SELECT MAX(salary) FROM employees) finds the highest salary.

. The outer query then finds the maximum salary less than that, which is the second highest.

Tip: To handle ties or retrieve multiple employees with the same second-highest salary, you can also use DENSE_RANK() with a CTE.

2) How do you use GROUP BY and HAVING together? Provide an example.

Answer:

Use GROUP BY to group data and HAVING to filter aggregated results (unlike WHERE, which filters raw rows).

```
SELECT department_id, COUNT(*) AS emp_count  
FROM employees  
GROUP BY department_id  
HAVING COUNT(*) > 5;
```

Explanation:

- . Groups employees by department.
- . Filters groups where the count of employees is more than 5.

3) Write a query to fetch the top 3 performing products based on sales.**Answer:**

Assume table sales_data has:

```
product_id, product_name, total_sales  
SELECT product_id, product_name  
, total_sales FROM sales_data ORDER BY total_sales DESC  
LIMIT 3;
```

Alternate using RANK((if ties matter):

```
SELECT product_id, product_name, total_sales  
FROM (  
SELECT *, RANK() OVER (ORDER BY total_sales DESC) AS rank_num  
FROM sales_data  
) ranked_sales  
WHERE rank_num <= 3;
```

4. How do you use a CASE statement in SQL? Provide an**Answer:**

Example.

CASE lets you write conditional logic in SQL (similar to IF/ELSE).

```
SELECT name, salary,
```

```
CASE
```

```
WHEN salary >= 100000 THEN 'High'
WHEN salary >= 50000 THEN 'Medium'
ELSE 'Low'
END AS salary_category
FROM employees;
```

Explanation:

- . Assigns a category based on salary value.
- . Works inside SELECT, WHERE, ORDER BY, etc.

5) Explain the difference between OLTP and OLAP systems with examples.

Answer:

OLTP (Online Transaction Processing) is optimized for transactional tasks like inserts, updates, and deletes. Example: an e-commerce website recording every order. OLAP (Online Analytical Processing) is designed for heavy read queries and aggregations, e.g., generating sales dashboards in Power BI. In one project, OLTP stored raw order data in MySQL, and OLAP stored transformed data in Snowflake for analytics. Tips: Always add a real project example. Mention that data engineers usually move data from OLTP → OLAP via ETL/ELT pipelines.

6) What's the difference between ETL and ELT? Which one do you prefer?

ETL (Extract-Transform-Load): Transformation happens before loading.

Example: Informatica, SSIS.

ELT (Extract-Load-Transform): Load raw data first, then transform inside the warehouse. Example: Snowflake, BigQuery.

In my project, we used ELT in Snowflake because compute was scalable.

```
WHEN salary >= 100000 THEN 'High'
WHEN salary >= 50000 THEN 'Medium'
ELSE 'Low'
END AS salary_category
FROM employees;
```

Explanation:

- . Assigns a category based on salary value.
- . Works inside SELECT, WHERE, ORDER BY, etc.

5) Explain the difference between OLTP and OLAP systems with examples.

Answer:

OLTP (Online Transaction Processing) is optimized for transactional tasks like inserts, updates, and deletes. Example: an e-commerce website recording every order. OLAP (Online Analytical Processing) is designed for heavy read queries and aggregations, e.g., generating sales dashboards in Power BI. In one project, OLTP stored raw order data in MySQL, and OLAP stored transformed data in Snowflake for analytics. Tips: Always add a real project example. Mention that data engineers usually move data from OLTP → OLAP via ETL/ELT pipelines.

6) What's the difference between ETL and ELT? Which one do you prefer?

ETL (Extract-Transform-Load): Transformation happens before loading.

Example: Informatica, SSIS.

ELT (Extract-Load-Transform): Load raw data first, then transform inside the warehouse. Example: Snowflake, BigQuery.

In my project, we used ELT in Snowflake because compute was scalable.

Tips:

Say ELT is more popular in modern cloud platforms. Mention real tools like Snowflake or Synapse.

7) How do you optimize a SQL query that runs slow on millions of records?

Use EXPLAIN plan. Add indexes. Avoid SELECT *. Use partitioning & clustering.

Example: Reduced query runtime from 20 mins to 2 mins by partitioning sales table by date and optimizing joins.

Tips:

Always give a before vs after performance improvement. Avoid textbook answers

8) What are common challenges in data pipelines? How do you handle them?

Answer:

Data quality issues → applied validation rules. Late-arriving data → handled via watermarking in ADF. Pipeline failures → retry + alerts in Airflow.

Example:

IoT logs ingestion required retry logic due to API failures.

Tips:

Show that you handle real-world messiness. Mention monitoring & alerting (CloudWatch, Airflow, ADF).